MID-STATE TECHNICAL COLLEGE · IT PROGRAMS

Network Storage and Backup Infrastructure

Unit 9 · Virtualization and SDDC (10-150-130)

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Learning Outcomes

By the end of this unit, you will be able to:



Select a network storage protocol

Choose NFS, iSCSI, or SMB for a workload and justify the choice.



Connect Proxmox VE to shared storage

Attach TrueNAS NFS and iSCSI storage to a cluster and move VM disks between storages.



Explain how Proxmox Backup Server protects data

Describe chunking, deduplication, incremental backups, encryption, and verification.



Design a backup strategy

Set RPO/RTO targets, retention, and tiers that survive hardware loss and ransomware.



LESSON 1

Networked Storage Protocols

File-Level vs. Block-Level Storage

File level — NFS, SMB

- The **server owns the filesystem**; clients request files by path.
- Many clients can read and write the same share at once.
- Proxmox stores disks as files — `.qcow2` or `.raw` — plus ISOs, templates, and backups.

Block level — iSCSI, FC, NVMe-oF

- The **client owns the filesystem**; the server only exposes raw blocks (a **LUN**).
- The host sees what looks like a local disk.
- Lower overhead for VM disks and databases; content limited to disk images.

Rule of thumb: if two hosts must format the same block device, something has to coordinate them — in Proxmox that is shared LVM or a cluster filesystem.

NFS — Network File System

- RPC-based file sharing native to Linux/Unix; **NFSv3** is stateless, **NFSv4.x** is stateful with a single port (2049).
- Client-side caching makes large sequential I/O — ISOs, backups — perform well.
- Proxmox treats NFS as **shared storage** that accepts every content type: images, containers, ISOs, templates, backups, snippets.
- Access is controlled by the server's **export list** — which hosts may mount, and read-only or read-write.



In our lab

The ISO library is the NFS store `VMLab-NFS`, mounted on every host from the helper node.

Proxmox mounts it under `/mnt/pve/<storage-id>`.

iSCSI — Block Storage over IP

- Carries SCSI commands over TCP (port 3260). The host is the **initiator**; the SAN is the **target**.
- Targets and initiators are named by **IQN** — `iqn.2005-10.org.freenas.ctl:proxmox`.
- Each target exposes one or more **LUNs** — on TrueNAS, a LUN is backed by a ZFS **zvol**.
- Proxmox content: disk images only. Layer **shared LVM** on the LUN so every node can use it.



Lab tuning

Create the zvol with a **4K block size** — not the TrueNAS default of 16K — for Proxmox VM disks.

Run iSCSI on the dedicated storage network with **MTU 9000** end to end.

SMB/CIFS — Windows File Sharing

- File-level protocol (TCP 445) built for **Windows** environments, with Active Directory user authentication.
- **SMB 3.x** adds per-share encryption, signing, and multichannel across several NICs.
- Proxmox can mount SMB as `cifs` storage for ISOs, templates, and backups; credentials are stored on each node.
- Hyper-V can run VMs directly from SMB 3 shares — we return to this in the Hyper-V storage unit.



Retire SMB1

"CIFS" names the original SMB1 dialect. It lacks encryption and was the spread path for WannaCry — disable it and require SMB 3.

Protocol Comparison

Most environments run more than one protocol — each where it fits best.

Protocol	Level	Best use case	Performance	Security
NFS	File	Linux shares, ISO libraries, backup targets	Strong for large sequential I/O	Host-based exports; Kerberos in v4
iSCSI	Block	VM disks, databases	Low latency, high IOPS	CHAP, isolated VLAN, IPsec
SMB 3	File	Windows shares, Hyper-V storage	Good for mixed workloads; multichannel	AD auth, signing, encryption

Pattern in practice: NFS for ISOs and backups, iSCSI for VM disks, SMB for Windows users and Hyper-V.

What Proxmox VE Can Put Where

Storage type determines content, sharing, and snapshot support — `pvesm status` shows what each node sees.

Storage type	Level	Content	Shared	Snapshots
NFS	File	All types	Yes	Yes, with qcow2
SMB/CIFS	File	All types	Yes	Yes, with qcow2
iSCSI (direct LUN)	Block	Disk images	Yes	No
LVM on iSCSI	Block	Disk images, containers	Yes	New in PVE 9 (volume chains)
ZFS over iSCSI	Block	Disk images, containers	Yes	Yes

LVM volume-chain snapshots arrived in PVE 9.0 as a technology preview — check the release notes before relying on them in production.

Securing Storage Traffic

- **Isolate** storage on its own VLAN or physical network — never routed to user or internet segments.
- **NFS**: restrict exports to the hosts that need them; NFSv3 trusts the client's IP and user IDs.
- **iSCSI**: use **CHAP** and restrict targets to known initiator IQNs.
- **SMB**: require SMB 3 signing or encryption and disable SMB1.



SECURITY FOCUS

Block storage has no file permissions. Any initiator that logs into a LUN can read — or overwrite — every VM disk on it.

Why it matters: One misconfigured host on the storage network can corrupt or exfiltrate an entire cluster's data.

LESSON 2

SANs and Shared Storage Management



Storage Area Network Architecture

- A **SAN** is a dedicated network that gives servers **block-level** access to consolidated storage.
- **Host adapters** (HBAs or NICs) connect servers to the **fabric** of SAN switches.
- **Storage arrays** hold the disks and present LUNs, with snapshots and replication built in.
- Redundant paths — **multipathing** — remove single points of failure between host and array.



Our lab SAN

Each POD: Host1–Host3 plus a TrueNAS SAN node.

Storage traffic rides its own network, referenced by IP — not DNS.

SAN Transports

The SAN concept is constant; the transport underneath it varies with budget and performance needs.

Transport	Runs over	Strength	Typical home
Fibre Channel	Dedicated FC fabric	Lossless, very low latency	Large enterprise arrays
iSCSI	Standard Ethernet / TCP	Low cost, familiar tools	SMB to mid-size, labs
FCoE	Lossless (DCB) Ethernet	Converges FC onto Ethernet	Legacy converged designs
NVMe-oF	TCP, RDMA, or FC	Lowest overhead for flash	Modern all-flash SANs
InfiniBand	IB fabric with RDMA	Ultra-low latency	HPC and AI clusters

TrueNAS 25.10 "Goldeye" added NVMe-oF targets alongside iSCSI — a sign of where block storage is heading.

Jumbo Frames on the Storage Network

Every hop must agree on MTU 9000 — the host NIC, the Linux bridge, the switch port, and TrueNAS.

```
ip link show vmbr1 | grep mtu
ping -M do -s 8972 <SAN-storage-IP>
```

- `-M do` forbids fragmentation; `8972` is 9000 minus 28 bytes of IP and ICMP headers.
- If the ping fails but a normal ping works, some hop is still at 1500.
- A mismatched MTU often passes small I/O and stalls on large transfers — the hardest kind of fault to spot.

Live Storage Migration

- **Storage migration** moves a running VM's disk to a different storage with no downtime.
- QEMU mirrors the disk in the background, then switches over once source and target are in sync.
- Plan for bandwidth and I/O load — the copy competes with production traffic.
- Containers: stop first, then use `pct move-volume`.

From the CLI

```
qm disk move 101 scsi0 SAN-LVM
```

Add `--delete 1` to remove the source; otherwise it stays as an unused volume.

Performance and Capacity Planning

Make it fast

- Dedicated storage network at **10 GbE or faster**; RDMA where supported.
- SSDs for random I/O; **tiering** and **caching** for hot data.
- Match the workload: databases want low-latency block, file shares want sequential throughput.

Keep it growing

- Track growth rate, not just current use — and include snapshots and backups.
- Expand non-disruptively: grow the zvol, then the LUN and LVM; ZFS 2.3+ supports **RAIDZ expansion**.
- Alert well before **80% pool usage** — ZFS performance drops as pools fill.

Remember: thin provisioning lets you promise more space than you own — monitoring is what keeps that promise honest.

LESSON 3

Proxmox Backup Server



What Proxmox Backup Server Is

- A dedicated, open-source backup server for VMs, containers, and physical Linux hosts.
- Added to Proxmox VE as a `pbs` storage; backup jobs target it like any other storage.
- **Incremental**, **deduplicated**, and **compressed** (zstd) — every backup is a full restore point.
- Built-in prune, garbage collection, verification, sync, and tape support.



Current release

PBS 4.2 (April 2026) on Debian 13 with ZFS 2.4.

Adds server-side encryption in sync jobs and S3-compatible object storage backends.

How Deduplication Works

Data is split into chunks; each chunk is named by its SHA-256 hash and stored only once per datastore.

Backup type	Chunking	How boundaries are set	Index file
VM disk images	Fixed size, ~4 MiB	Every 4 MiB of the disk	`.fidx`
Containers, hosts	Dynamic size	Rolling hash over a pxar archive	`.didx`

- A snapshot is just an **index** — a list of chunk hashes. Many snapshots share the same chunks.
- The client checks the last snapshot's index and uploads only chunks the server doesn't already have.

Incremental Backups with Dirty Bitmaps

- QEMU tracks which blocks changed since the last backup in a **dirty bitmap**.
- The next backup reads and sends only those blocks — no full disk scan.
- Every backup is still a complete restore point; there is no chain of incrementals to replay.



When the bitmap resets

Shutting a VM down, or rebooting its host, discards the bitmap. The next backup reads the whole disk — slower, but only new chunks are uploaded.

Logical Size vs. Space on Disk

The backup list shows how much a restore needs; the datastore summary shows what is really stored.

What the snapshot shows

- The full logical size of the guest — what a restore will write.
- Example: a 92 GB VM appears as 92 GB in every snapshot.

What the datastore uses

- Only unique chunks, compressed — the same VM might occupy 15 GB.
- The **deduplication factor** in the datastore summary shows the ratio.

Plan conservatively: dedup ratios depend on the data. Cloned VMs share most chunks; encrypted or already-compressed data barely dedupes.

Datastore Maintenance Jobs

Four scheduled jobs keep a datastore healthy — and explain why freed space appears a day later.

Job	What it does	Why it matters
Prune	Removes snapshots outside the retention policy (keep-daily, keep-weekly…)	Enforces retention; does not free space by itself
Garbage collection	Marks chunks still referenced, then deletes the rest a day later	Actually reclaims disk space
Verify	Re-reads chunks and checks their checksums	Catches bit rot before a restore does
Sync	Copies snapshots to or from another PBS (pull or push)	Puts a copy off-site

PBS 4.x can also back a datastore with S3-compatible object storage — introduced as a technology preview in 4.0.

Encryption and Integrity

- **Client-side AES-256-GCM** encryption — data is encrypted before it leaves the Proxmox host.
- All traffic between client and server runs over **TLS**.
- Every chunk carries a checksum; **verify jobs** prove backups are still readable.
- Back up the key itself: `proxmox-backup-client key paperkey` prints a recovery copy.



SECURITY FOCUS

An encrypted backup is only as recoverable as its key. Store the key away from the hosts it protects.

Why it matters: If the key lives only on the Proxmox host and that host is lost, every encrypted backup is lost with it.

LESSON 4

Backup Infrastructure Design



RPO and RTO

Recovery Point Objective (RPO)

- How much **data** the business can afford to lose, measured in time.
- Nightly backups mean an RPO of up to 24 hours.
- Drives backup frequency and replication.

Recovery Time Objective (RTO)

- How long the business can be **down** before recovery must finish.
- A 4-hour RTO rules out restoring 5 TB over a slow link.
- Drives restore speed, local copies, and standby capacity.

Also design for: retention requirements set by policy or regulation, and the backup window — the time you have to finish each job.

The 3-2-1-1-0 Rule

A modern extension of 3-2-1 that assumes an attacker will go after your backups:



3 copies of the data

Production plus two backups.



2 different media · 1 off-site

Disk and tape or object storage; one copy at another location (a PBS sync target).



1 offline or immutable copy

Removable datastore, tape, or object lock — beyond the reach of compromised credentials.



0 errors

Verify jobs and scheduled test restores prove the backups actually work.

Multi-Tier Backup Architecture

Keep the newest backups closest to production; push older copies to cheaper, more distant media.

Tier	Media	Retention	Purpose
Primary	Local PBS datastore on SSD/HDD	Days to weeks	Fast restores of recent mistakes
Secondary	Remote PBS via sync job	Weeks to months	Site loss; ransomware recovery
Archive	Tape or S3 object storage	Months to years	Compliance and long-term retention

Hybrid in practice: restore locally for speed, replicate to the cloud or a second site for disaster recovery — and check data-sovereignty rules first.

Data Reduction: Dedup and Compression

- **Source-side deduplication** — the client skips chunks the server already holds, saving bandwidth.
- **Target-side deduplication** — the storage system removes duplicates after data arrives.
- **Zstandard (zstd)** compresses each chunk quickly enough to run inline during backup.
- Reduction ratios depend entirely on the data — measure, don't assume.



Order of operations

Encrypt after compressing — encrypted data looks random and won't compress or dedupe. PBS handles this order for you.

Security Across the Unit



SECURITY FOCUS

Storage: isolate storage networks, restrict NFS exports and iSCSI initiators, require CHAP and SMB 3 encryption.

Backups: encrypt client-side, keep the key off the hosts, and run PBS with its own credentials — not shared with Proxmox VE.

Resilience: keep one copy off-site and one offline or immutable; prove it with verify jobs and test restores.

Why it matters: Ransomware operators look for backups first. If they can reach your storage network or your backup server with the same credentials, they can delete the recovery before they demand the ransom.

Hands-On Lab: Networked Storage

NETLAB+ · Lab 9: Networked Storage · your POD's Host1–Host3 and TrueNAS SAN



- Verify MTU 9000 end to end on the storage network.
- Connect all three hosts to TrueNAS over NFS and iSCSI.
- Layer shared LVM on the iSCSI LUN and confirm every node sees it.
- Move a running VM's disk from local storage to the SAN.
- Document which content types each storage accepts, and why.

Graded against the live configuration on your POD — see Blackboard Ultra for the full assignment.

Key Takeaways

- File-level storage (NFS, SMB) shares files; block-level storage (iSCSI) shares raw disks.
- Shared SAN storage enables live migration and HA — and must be isolated and jumbo-frame clean.
- PBS stores each SHA-256 chunk once, so every incremental backup is a full restore point.
- RPO and RTO come from the business; tiers, retention, and frequency follow from them.
- A backup you haven’t verified and test-restored is a hope, not a backup.

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Questions?

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